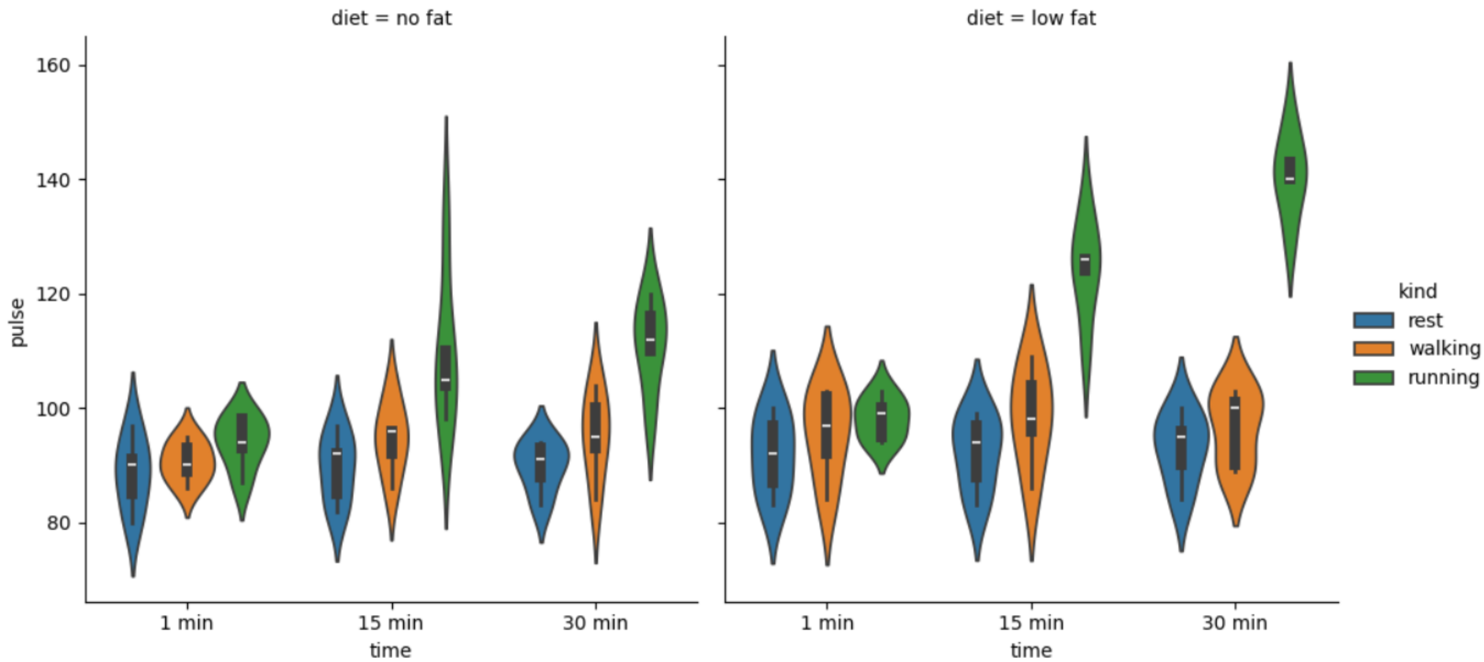
Exercise Pulse Rate Analysis Using Violin Plot

[26]:
```
import pandas as pd
import seaborn as sb
from matplotlib import pyplot as plt
df = sb.load_dataset('exercise')
sb.catplot(x = "time", y = "pulse", hue = "kind", kind = 'violin', col = "diet", data = df);
plt.show()
```



Introduction

This violin plot is used to analyze the pulse rate of people during different types of exercises at different time intervals and diet conditions.

The graph is created using the Seaborn catplot() function with kind='violin'.

Variables Used in the Plot

Variable	Meaning
time	Exercise duration (1 min, 15 min, 30 min)
pulse	Pulse rate of individuals
kind	Type of activity (rest, walking, running)
diet	Diet category (no fat, low fat)

Exercise Pulse Rate Analysis Using Violin Plot

Explanation of the Plot

The violin plot compares pulse rate distributions based on:

- Exercise time
- Exercise type
- Diet condition

The graph is divided into two sections:

1. **No Fat Diet**
2. **Low Fat Diet**

Different colors represent:

- Blue → Rest
- Orange → Walking
- Green → Running

Interpretation

1. Pulse Rate During Rest

- Pulse rate remains lower compared to other activities.
- Most pulse values are concentrated around 85–100.
- Rest activity shows less variation.

2. Pulse Rate During Walking

- Pulse rate increases slightly while walking.
- Most pulse values are between 90–105.
- Walking produces moderate pulse variation.

Exercise Pulse Rate Analysis Using Violin Plot

3. Pulse Rate During Running

- Running shows the highest pulse rate.
- Pulse values increase significantly at 15 and 30 minutes.
- Maximum pulse values reach around 140–150 in low-fat diet conditions.

4. Effect of Time

- As exercise time increases from 1 minute to 30 minutes:
    - Pulse rate gradually increases.
    - Running activity shows the largest increase.

5. Effect of Diet

- People under the low-fat diet show slightly higher pulse values during running.
- Pulse distribution differs between diet categories.

6. What Does the Width of the Violin Represent?

- Wider area → More observations concentrated in that pulse range.
- Narrow area → Fewer observations.

7. What Does the White Dot Represent?

The white dot inside the violin indicates the median pulse value.

Exercise Pulse Rate Analysis Using Violin Plot

8. What does the X-axis represent?

The X-axis represents exercise time intervals:

- 1 min
- 15 min
- 30 min

9. What does the Y-axis represent?

The Y-axis represents pulse rate values.

10. What does the hue variable represent?

The hue variable represents exercise type:

- Rest
- Walking
- Running

11. What does the col variable represent?

The col variable separates the graph based on diet category:

- No fat
- Low fat

12. Which activity produces the highest pulse rate?

Running produces the highest pulse rate.

13. How does pulse rate change with time?

Pulse rate increases as exercise duration increases.

Exercise Pulse Rate Analysis Using Violin Plot

14. Which diet condition shows higher pulse values during running?

The low-fat diet condition shows slightly higher pulse values during running.

15. Why is the running violin wider?

The running violin is wider because pulse values are more spread out and variable.

16. What is the purpose of a violin plot?

A violin plot is used to visualize:

- Distribution
- Density
- Spread
- Median of numerical data

17. What conclusion can be drawn from this graph?

The graph shows that pulse rate increases with exercise intensity and duration. Running causes the highest pulse rate, especially under low-fat diet conditions.

Conclusion

The violin plot clearly shows the relationship between exercise duration, activity type, diet condition, and pulse rate. Running activity and longer exercise duration significantly increase pulse rate compared to resting and walking activities.